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Coupled FSI Analysis of a Photovoltaic Structure Using Open Source Software

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Course: Open Source CAE Tools

agh.edu.pl

24.06.2025 r.

Aim of the Project

- Perform a one-way coupled FSI simulation using available open-source software:
  - CFD simulation – OpenFOAM
  - symulacja strukturalna – Calculix (w środowisku PrePoMax)
- Acquire proficiency in using and applying open-source software tools

Photovoltaic Structure and Associated Hazard





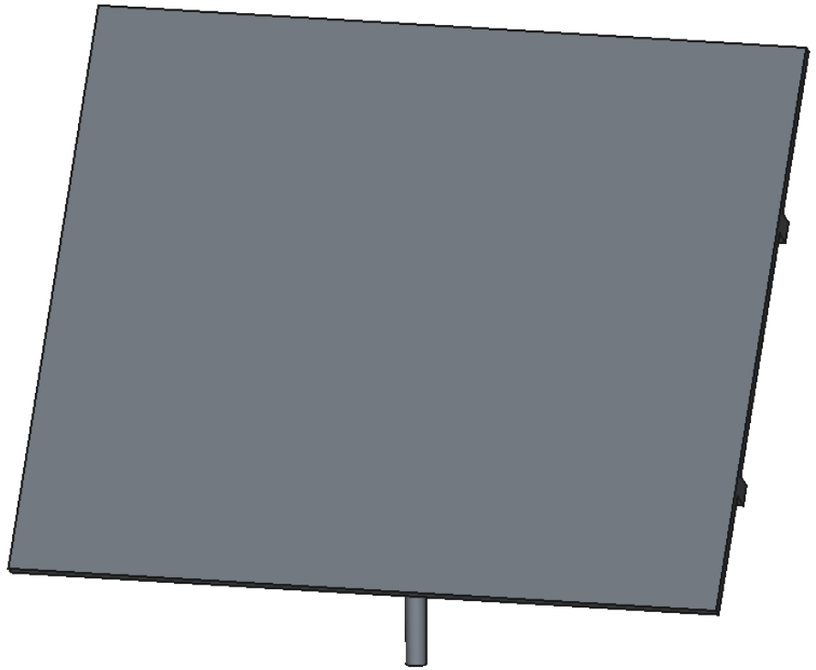
Maximum wind speed in Poland:

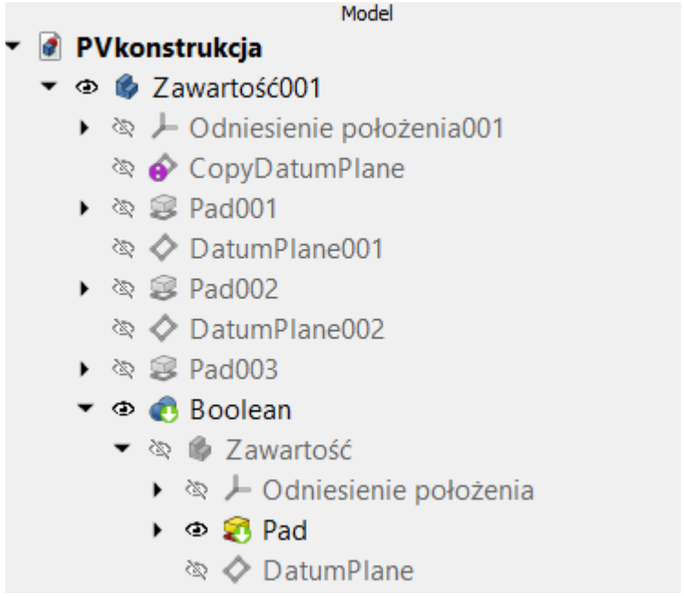
$$v = 47\frac{m}{s} \cong 169\frac{km}{h}$$

Software Used

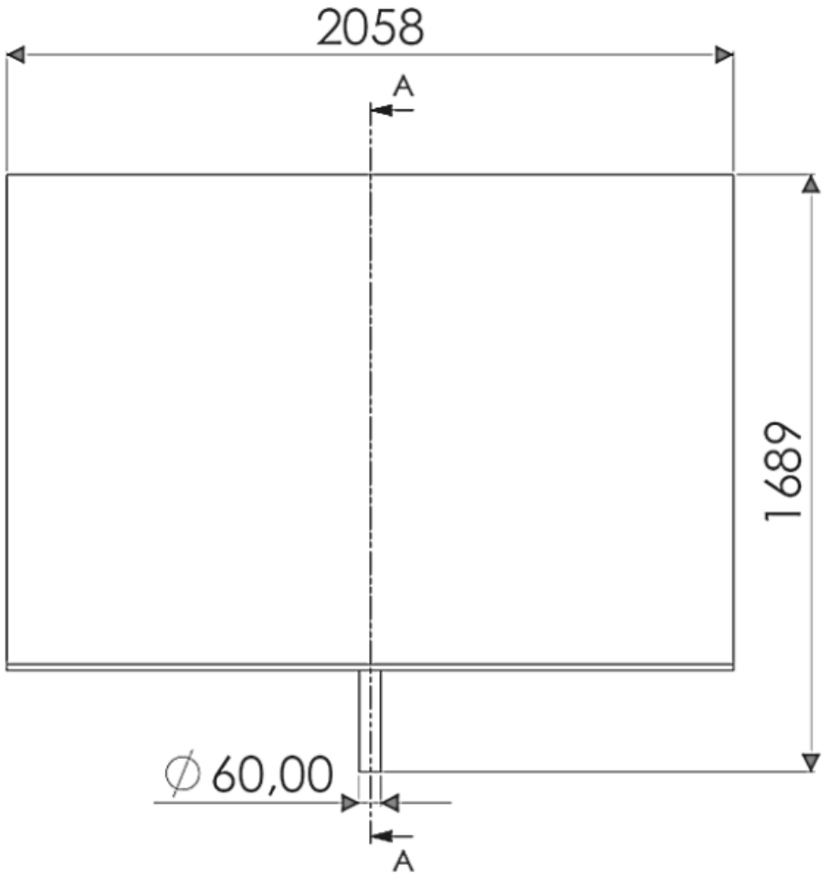
1. FreeCAD 1.0.1 (LGPL2+) – Windows 11
2. Notepad ++ v8.8.1 (GNU GPL) – Windows 11
3. Open Foam v2412 (GNU GPL) – Linux (Ubuntu 22.04.5 LTS)
4. ParaView 6.0.0 RC1 (BSD 3-Clause) – Windows 11
5. PrePoMax v2.3.5dev (GNU GPL) – Windows 11
   - CalculiX 2.22 (GNU GPL)

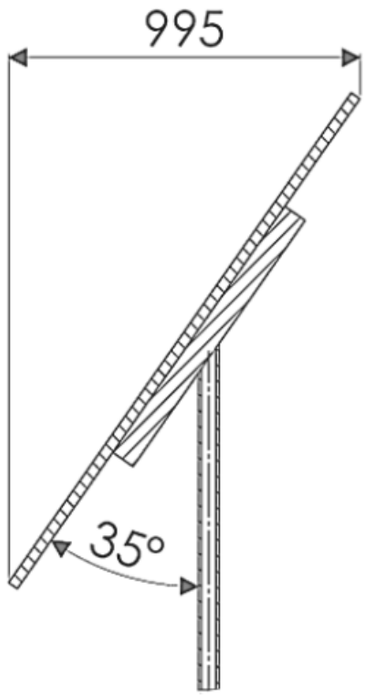
Solid model of the Structure





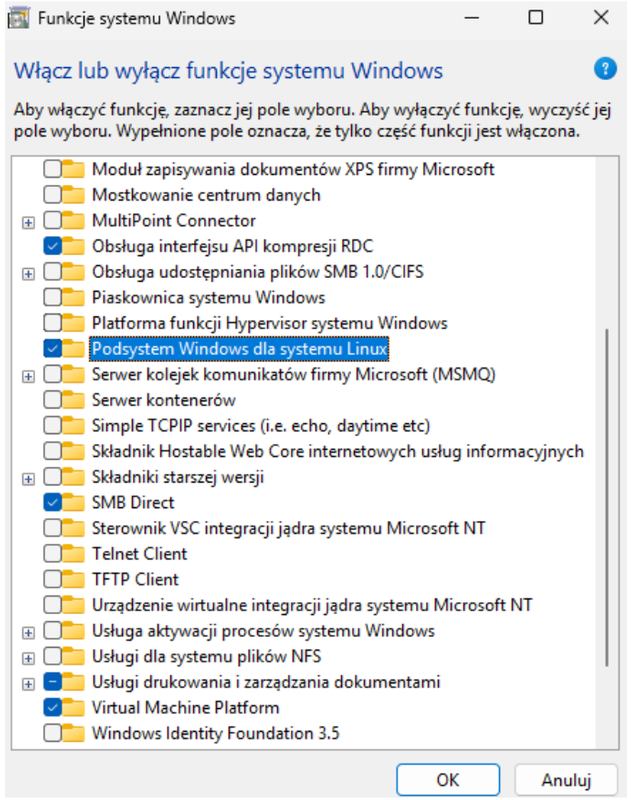
Solid model of the Structure





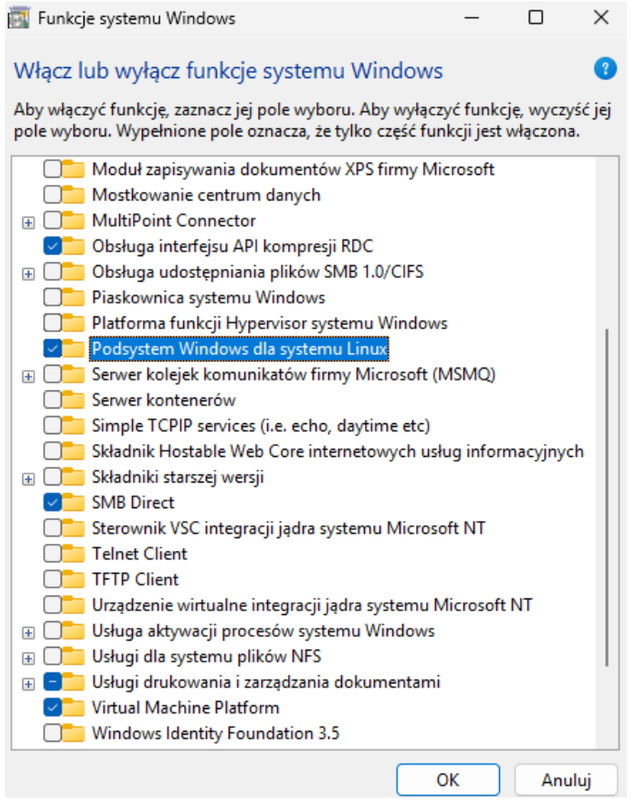
LINUX Environment Setup

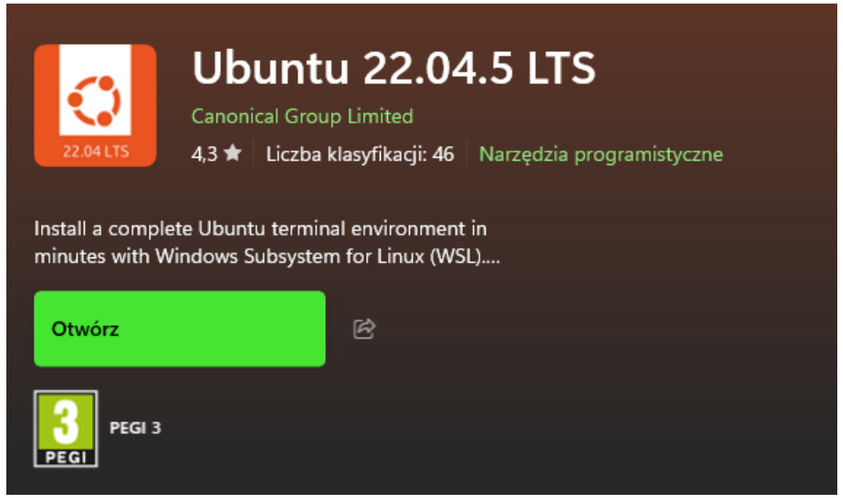
Creating a LINUX environment on Windows 11



LINUX Environment Setup

reating a LINUX environment on Windows 11





LINUX Environment Setup

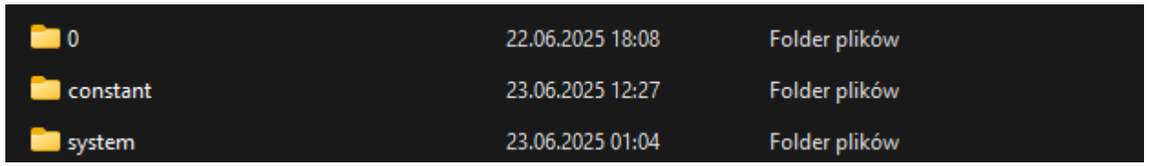
Installing OpenFOAM in the LINUX environment

Commands used:

```
sudo sh -c "wget -O - https://dl.openfoam.com/add-debian-repo.sh | bash„
sudo apt install openfoam2412
```

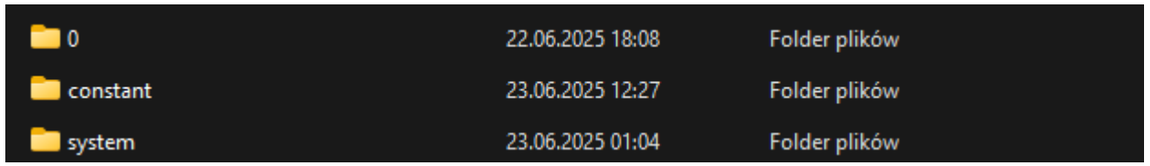
CFD Simulation

Creating the Folder Structure Required for the Simulation



CFD Simulation

Creating the Folder Structure Required for the Simulation

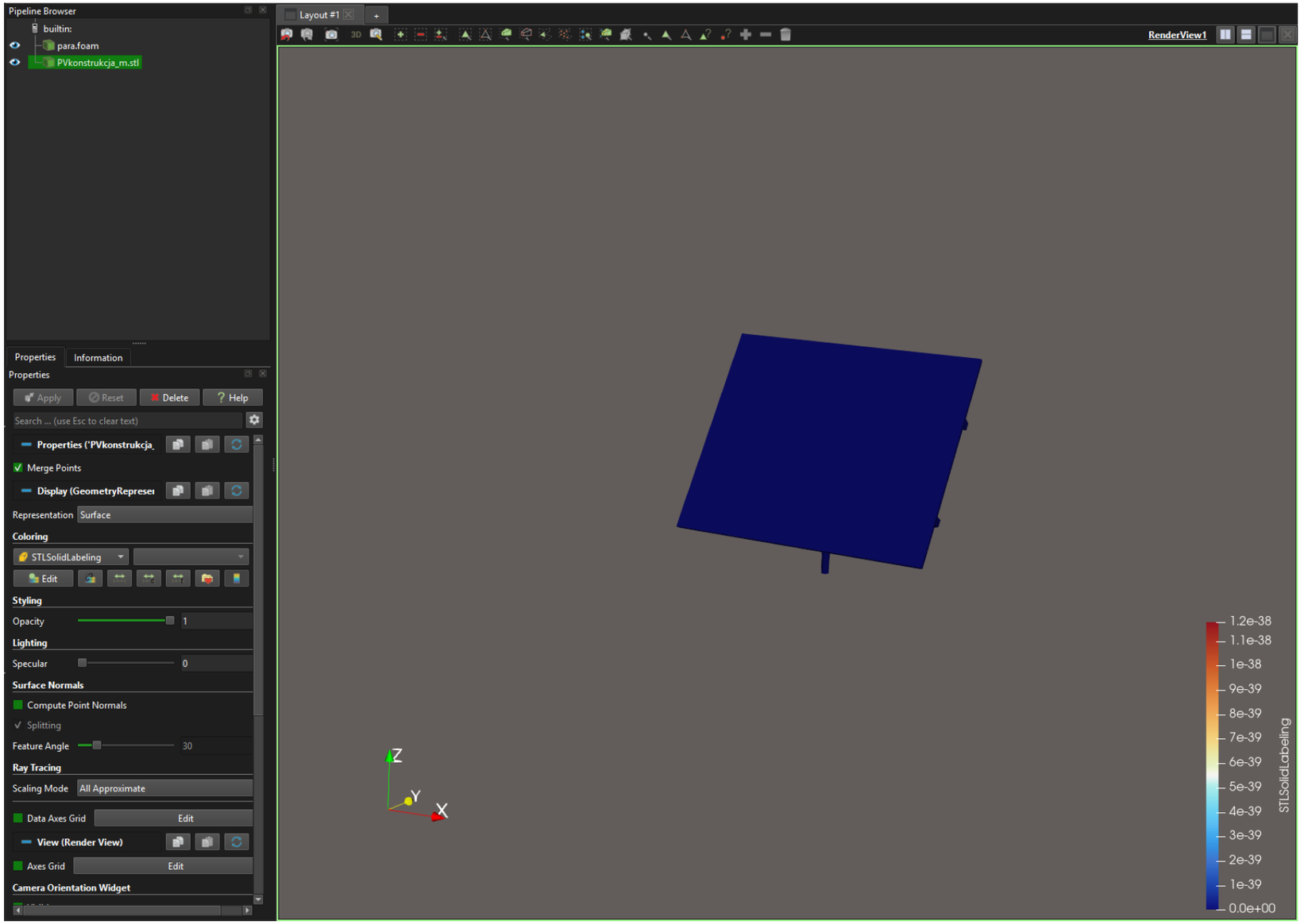


Saving the STL file in the `/constant/triSurface` stl file, then using a command to scale the geometry:

```
surfaceConvert PVkonstrukcja.stl PVkonstrukcja_m.stl -clean -scale 0.001
```

CFD Simulation

Scaled Model in the ParaView Environment



MESH Settings

1. blockMesh – Defining the Computational Domain

```
/*--------------------------------*- C++ -*----------------------------------*\
| =========                 |                                                 |
| \\      /  F ield         | OpenFOAM: The Open Source CFD Toolbox           |
|  \\    /   O peration     | Version:  2.2.1                                 |
|   \\  /    A nd           | Web:      www.OpenFOAM.org                      |
|    \\/     M anipulation  |                                                 |
\*---------------------------------------------------------------------------*/
FoamFile
{
    version     2.0;
    format      ascii;
    class       dictionary;
    object      blockMeshDict;
}
// * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * //

convertToMeters 1;  //This gives you the scale: 0.001 would mean all units in this file are [mm]

vertices        //list of vertices defining the geometry
(
//        X     Y     Z
( -3.0  -8.0   0.0 )   // 0: lewy-przod-dol
(  3.0  -8.0   0.0 )   // 1: prawy-przod-dol
(  3.0   12.0  0.0 )   // 2: prawy-tyl-dol
( -3.0   12.0  0.0 )   // 3: lewy-tyl-dol

( -3.0  -8.0   7.0 )   // 4: lewy-przod-gora
(  3.0  -8.0   7.0 )   // 5: prawy-przod-gora
(  3.0   12.0  7.0 )   // 6: prawy-tyl-gora
( -3.0   12.0  7.0 )   // 7: lewy-tyl-gora
);

blocks          //defining the block which the geometry is made out of
(
 hex (0 1 2 3 4 5 6 7) (30 100 40) simpleGrading (1 1 1)
//this means that the only block in the geometry is defined by vertices (0 1 2 ...), there are 10
//and 1 cells in the x, y, and z direction and that in each direction all the cells have the same
);

edges
(
//    spline 0 1
//    (
//      (5 -2 0)
//    )
//    spline 4 5
//    (
//      (5 -2 0.1)
//    )
```

```
boundary
(
    UpperandUnder   //choose a name for the boundary
    {
        type wall;  //define the type of the boundary
        faces
        (
            (1 2 3 0)   //list the faces; vertices should be listed in clockwise direction, as se
            (7 6 5 4)
        );
    }

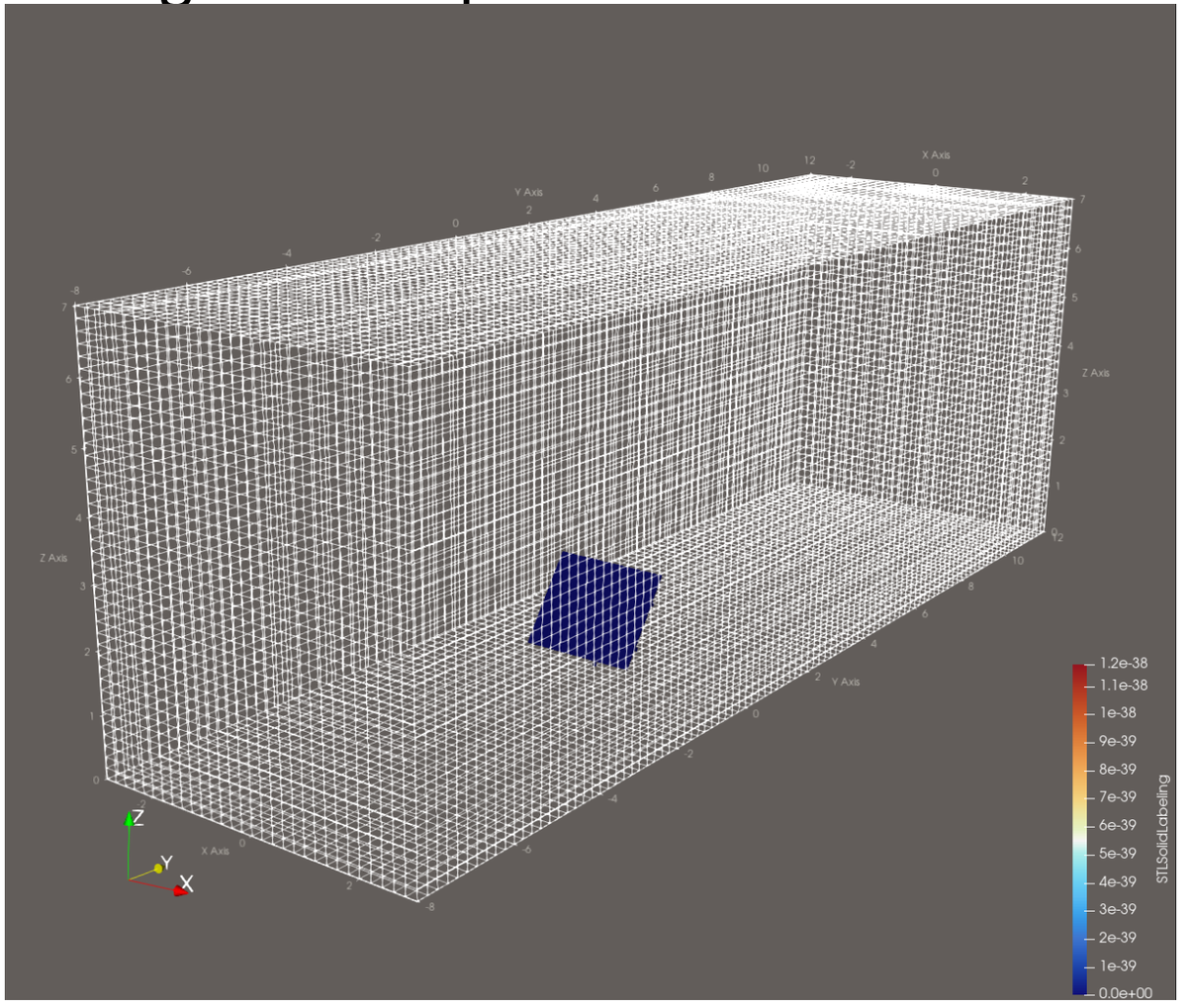

    inlet
    {
        type patch;
        faces
        (
            (0 4 5 1)
    );
    }


    outlet
    {
    type patch;
    faces
    (
            (2 6 7 3)
    );
    }


    frontAndBack
    {
        type wall;
        faces
        (
            (2 1 5 6)
            (3 7 4 0)
        );
    }
);
```

MESH Settings

1. blockMesh – Defining the Computational Domain



MESH Settings

2. surfaceFeatureExtract – STL Information for Further Mesh Generation

```
/*--------------------------------*- C++ -*----------------------------------*\
| =========                 |                                                 |
| \\      /  F ield         | OpenFOAM: The Open Source CFD Toolbox           |
|  \\    /   O peration     | Version:  2.2.1                                 |
|   \\  /    A nd           | Web:      www.OpenFOAM.org                      |
|    \\/     M anipulation  |                                                 |
\*---------------------------------------------------------------------------*/
FoamFile
{
    version     2.0;
    format      ascii;
    class       dictionary;
    object      surfaceFeatureExtractDict;
}
// * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * //

PVkonstrukcja_m.stl
{
    // How to obtain raw features (extractFromFile || extractFromSurface)
    extractionMethod    extractFromSurface;

    extractFromSurfaceCoeffs
    {
        // Mark edges whose adjacent surface normals are at an angle less
        // than includedAngle as features
        // - 0  : selects no edges
        // - 180: selects all edges
        includedAngle   150;
    }

    // Write options

        // Write features to obj format for postprocessing
        writeObj                yes;
}
```

MESH Settings

3. snappyHexMesh – Automatic generation of a hexagonal–prismatic mesh

```
/*--------------------------------*- C++ -*----------------------------------*\
| =========                 |                                                 |
| \\      /  F ield         | OpenFOAM: The Open Source CFD Toolbox           |
|  \\    /   O peration     | Version:  2.2.1                                 |
|   \\  /    A nd           | Web:      www.OpenFOAM.org                      |
|    \\/     M anipulation  |                                                 |
\*---------------------------------------------------------------------------*/
FoamFile
{
    version     2.0;
    format      ascii;
    class       dictionary;
    object      snappyHexMeshDict;
}

// * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * //

// Which of the steps to run
castellatedMesh true;
snap            true;
addLayers       false;


// Geometry. Definition of all surfaces. All surfaces are of class
// searchableSurface.
// Surfaces are used
// - to specify refinement for any mesh cell intersecting it
// - to specify refinement for any mesh cell inside/outside/near
// - to 'snap' the mesh boundary to the surface
geometry
{
    PVkonstrukcja_m.stl
    {
        type triSurfaceMesh;
        name PVkonstrukcja_m;
    }

    refinementBox
    {
        type box;
        min (-1.529 -1.038491 -0.5);
        max (2 2 3);
    }

};
```

```
// Explicit feature edge refinement
// ~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~

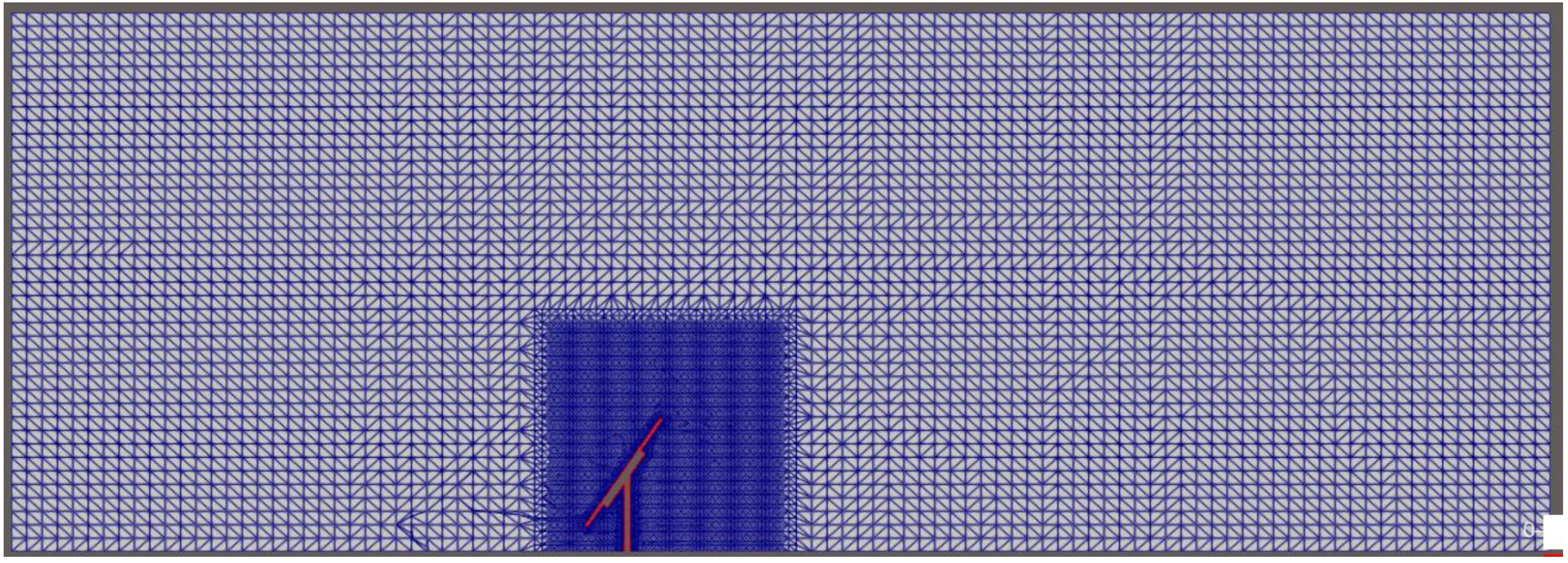
// Specifies a level for any cell intersected by its edges.
// This is a featureEdgeMesh, read from constant/triSurface for now.
features
(
    {
        file "PVkonstrukcja_m.eMesh";
        level 2; //1
    }
);
```

```
refinementSurfaces
{
    PVkonstrukcja_m
    {
        // Surface-wise min and max refinement level
        level (3 5); //1-5

        patchinfo
        {
            type wall
            inGroups (PVkonstrukcja_m);
        };
    }
};
```

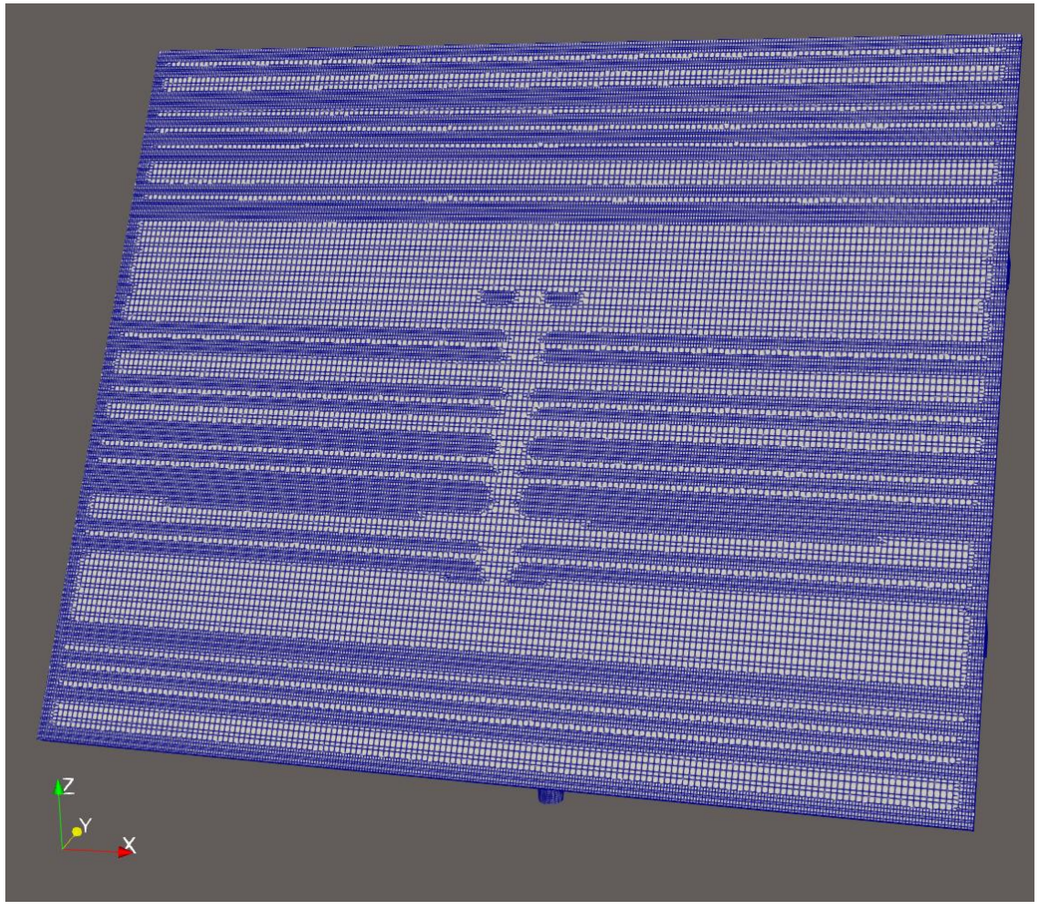
MESH Settings

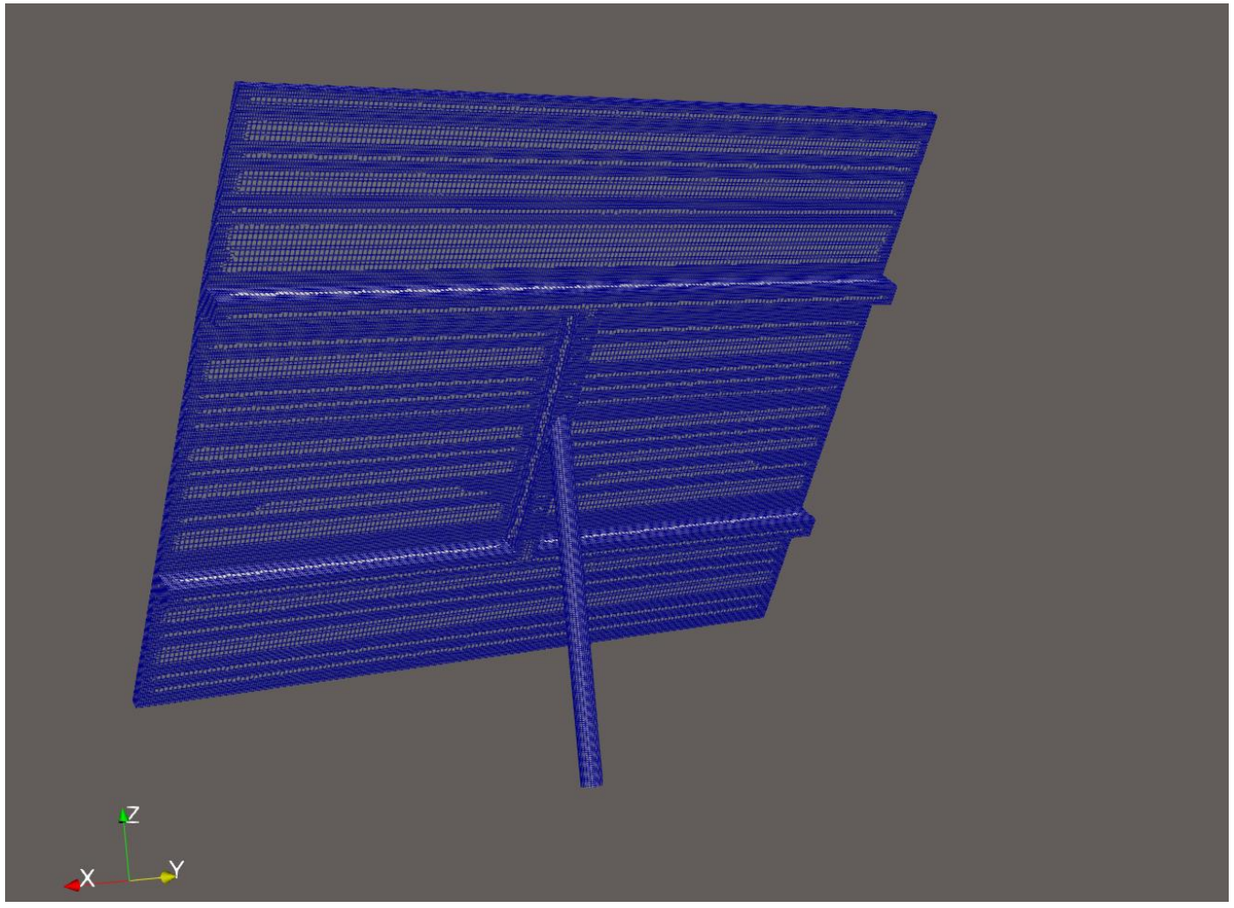
3. snappyHexMesh – Automatic generation of a hexagonal–prismatic mesh



MESH Settings

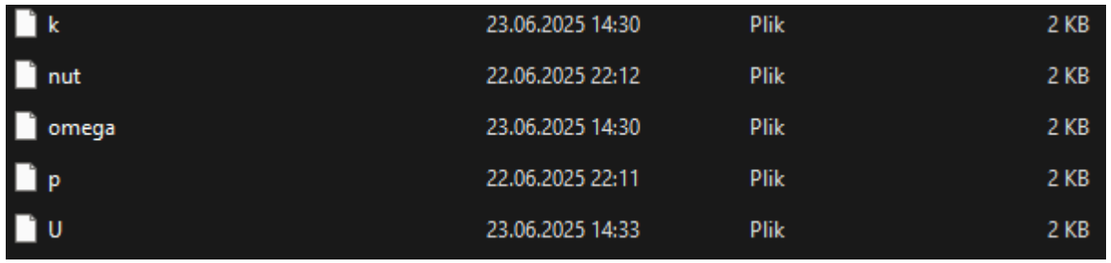
3. snappyHexMesh – Automatic generation of a hexagonal–prismatic mesh





Boundary Conditions and Simulation Files

- Simulation parameter settings for each variable using the **k–ω SST** turbulence model



Internal Field:
Turbulent kinetic energy k: 4.15
Turbulent viscosity: 0 m^2/s
Specific dissipation rate: 15.5 1/s

Boundary Conditions and Simulation Files

- Simulation parameter settings for each variable using the **k–ω SST** turbulence model

```
internalField   uniform 0;

boundaryField
{
    UpperandUnder
    {
        type            zeroGradient;
    }

    inlet
    {
        type            zeroGradient;
    }

    outlet
    {
        type            fixedValue;
    value       $internalField;
    }

    frontAndBack
    {
        type            zeroGradient;
    }
    PVkonstrukcja_m
    {
        type            zeroGradient;
    }

}

// ************************************************************************* //
```

```
internalField   uniform (0 0 0);

boundaryField
{
    UpperandUnder
    {
        type            fixedValue;
        value           uniform (0 0 0);
    }

    inlet
    {
        type            fixedValue;
        value           uniform (0 47 0);
    }

    outlet
    {
        type            inletOutlet;
    inletValue  uniform (0 0 0);
    value       $internalField;
    }

    frontAndBack
    {
        type            fixedValue;
        value           uniform (0 0 0);
    }
    PVkonstrukcja_m
    {
        type            fixedValue;
      value         uniform (0 0 0);
    }

}
```

Solver files

DecomposerPar – File for Multicore Simulation Setup

```
/*--------------------------------*- C++ -*----------------------------------*\
| =========                 |                                                 |
| \\      /  F ield         | OpenFOAM: The Open Source CFD Toolbox           |
|  \\    /   O peration     | Version:  4.x                                   |
|   \\  /    A nd           | Web:      www.OpenFOAM.org                      |
|    \\/     M anipulation  |                                                 |
\*---------------------------------------------------------------------------*/
FoamFile
{
    version     2.0;
    format      ascii;
    class       dictionary;
    note        "mesh decomposition control dictionary";
    object      decomposeParDict;
}
// * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * //

numberOfSubdomains  8;

method          scotch;
```

Solver files

Control – File defining the simulation parameters

```
/*--------------------------------*- C++ -*----------------------------------*\
| =========                 |                                                 |
| \\      /  F ield         | OpenFOAM: The Open Source CFD Toolbox           |
|  \\    /   O peration     | Version:  5                                     |
|   \\  /    A nd           | Web:      www.OpenFOAM.org                      |
|    \\/     M anipulation  |                                                 |
\*---------------------------------------------------------------------------*/
FoamFile
{
    version     2.0;
    format      ascii;
    class       dictionary;
    location    "system";
    object      controlDict;
}
// * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * //

application     simpleFoam;

startFrom       startTime;

startTime       0;

stopAt          endTime;

endTime         1000;

deltaT          1;

writeControl    timeStep;

writeInterval   100;

purgeWrite      0;

writeFormat     ascii;
```

```
writePrecision  6;

writeCompression off;

timeFormat      general;

timePrecision   6;

runTimeModifiable true;

functions
{
    samplePatchPressure
    {
        type              surfaceFieldValue;
        libs              ("libfieldFunctionObjects.so");
        log               true;
        writeControl      timeStep;
        writeInterval     100;
        writeFields       true;

        surfaceFormat      vtk;
        regionType         patch;
        name               PVkonstrukcja_m;

        operation          none;

        fields
        (
            p
        );
    }
}
```

Running Simulation

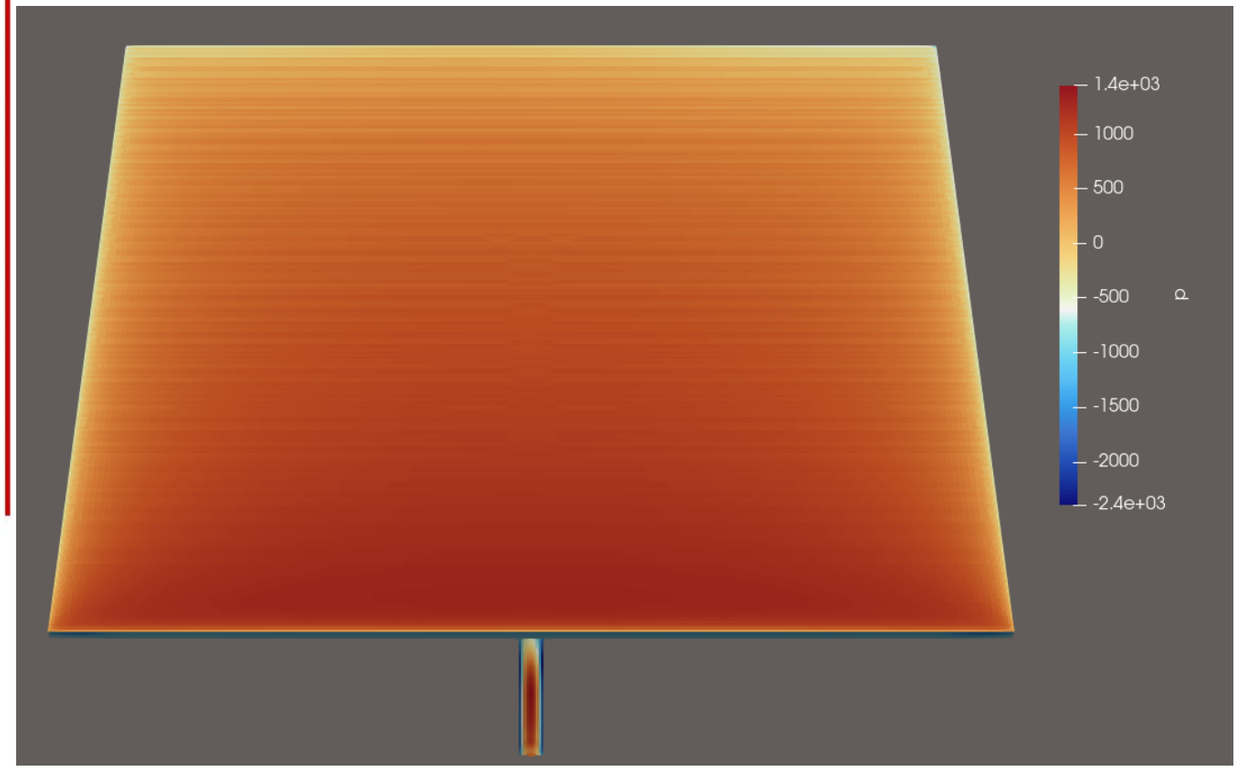
1. Docomposition
`decomposePar` – splits the case for paralel processing

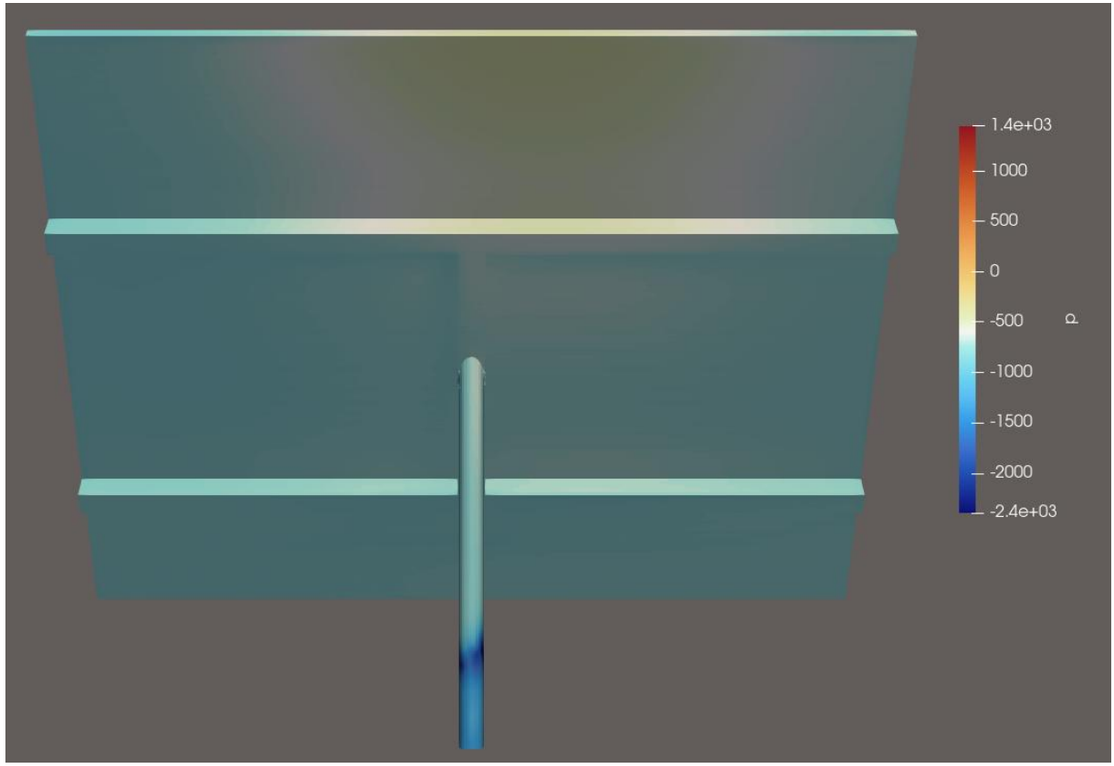
2. Multicore Simulation Execution
```
mpirun -np 8 simpleFoam –parallel
```

3. Results Reconstruction
`reconstructPar`

Results

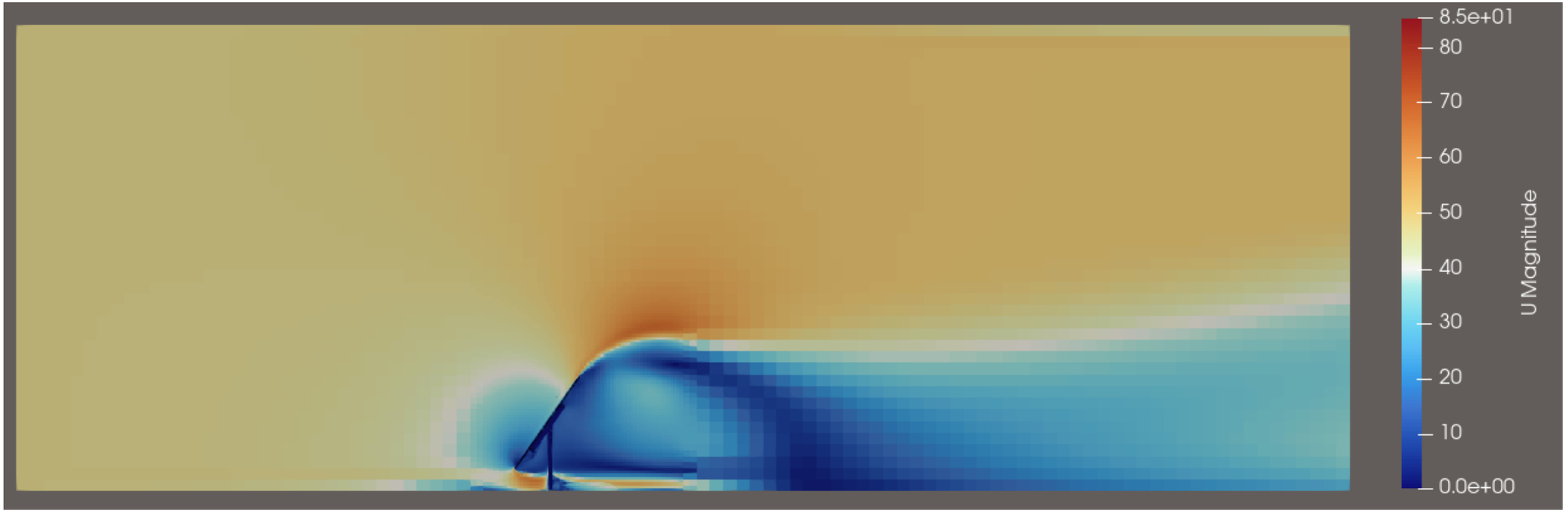
Static Pressure on the Surface of the Structure





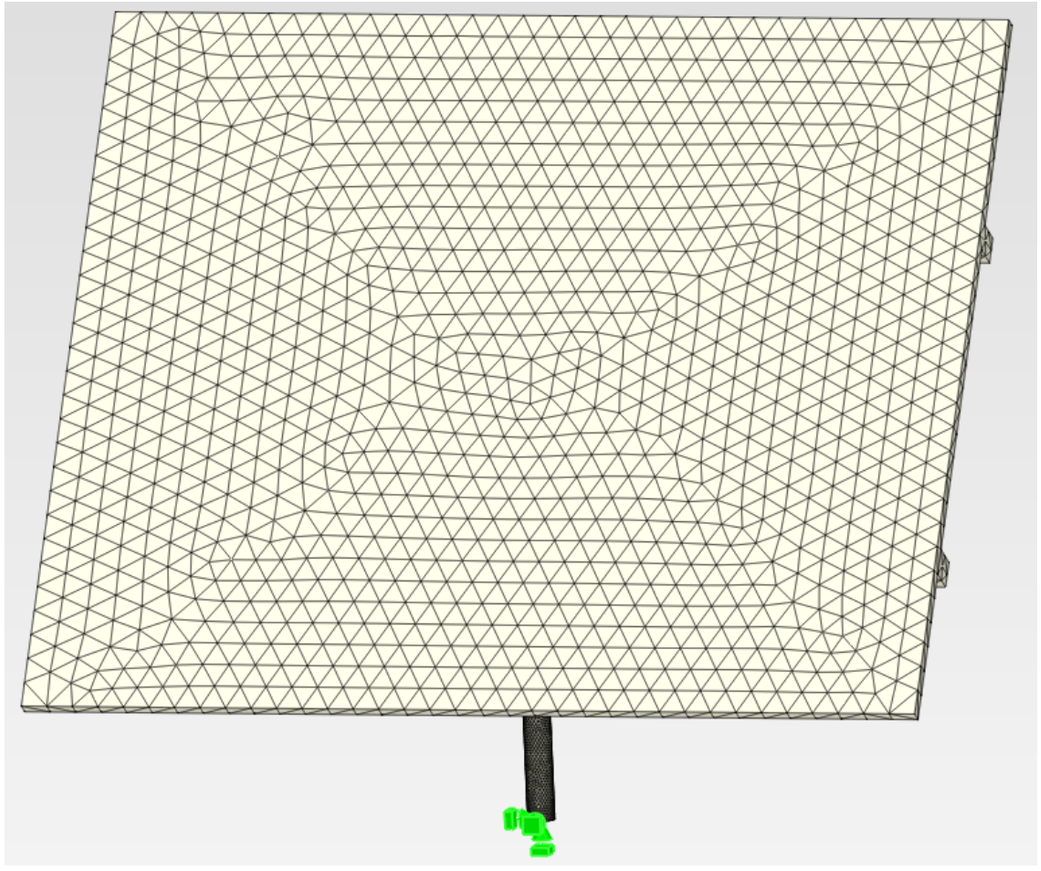
Results

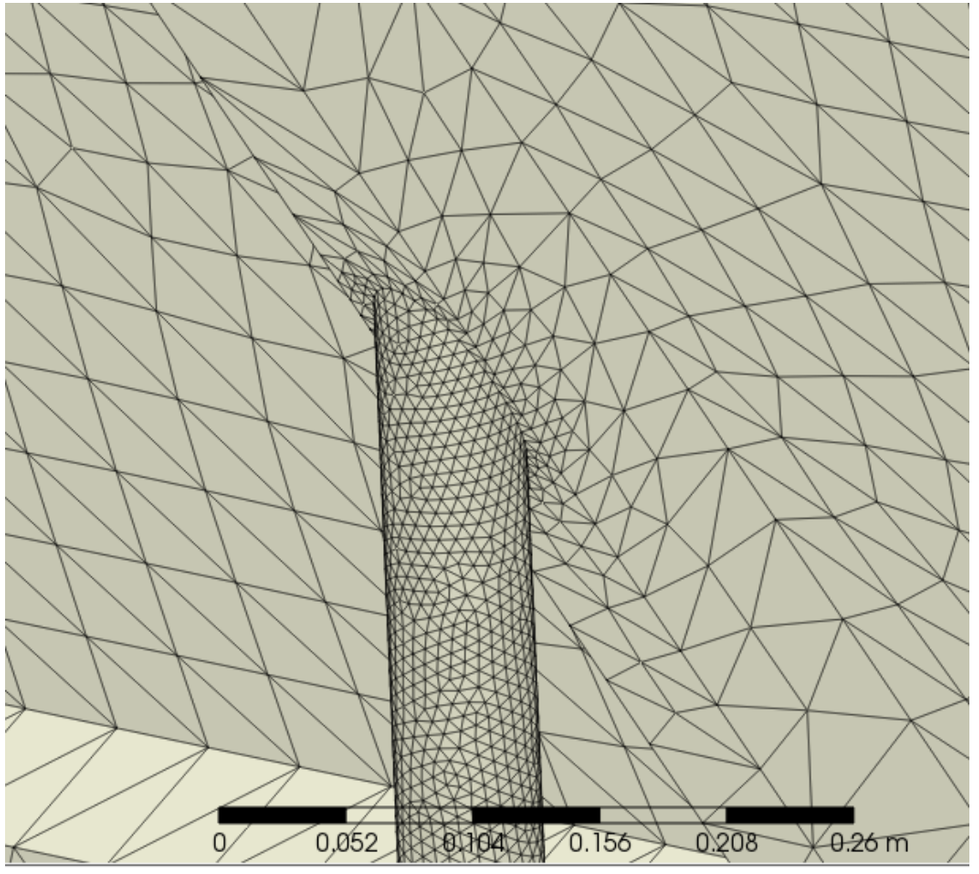
Velocity



Structural FEM Simulation

Discretized Model





Structural FEM Simulation

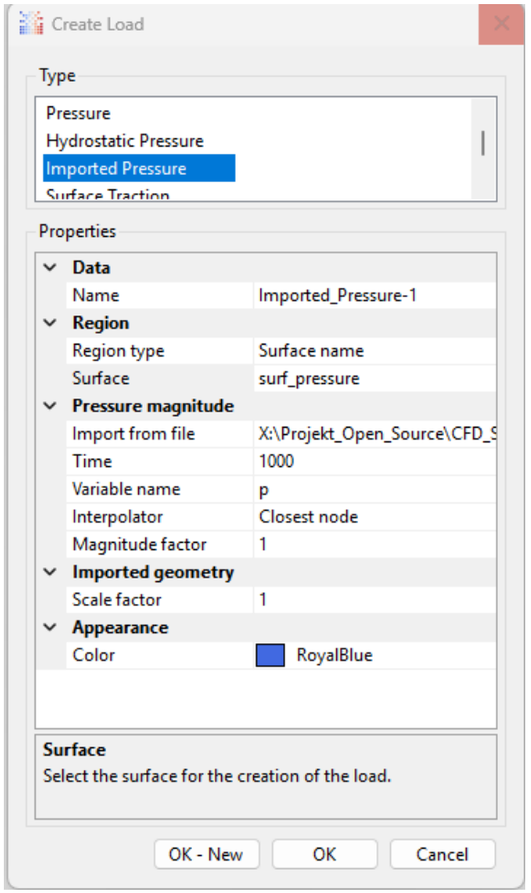
Material:
Structural steel

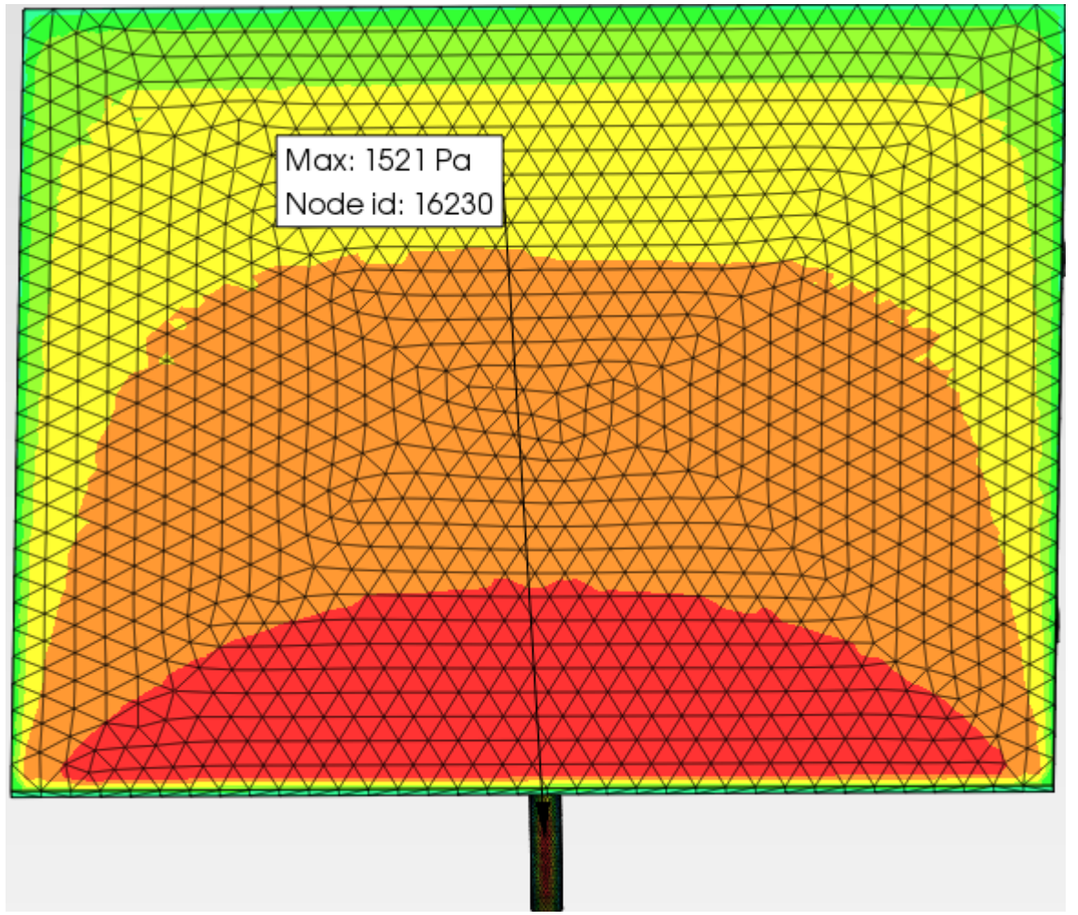
Parameters:

- Density – 7800 kg/m³
- Young’s modulus – 200 GPa
- Poisson’s ratio – 0.3

Structural FEM Simulation

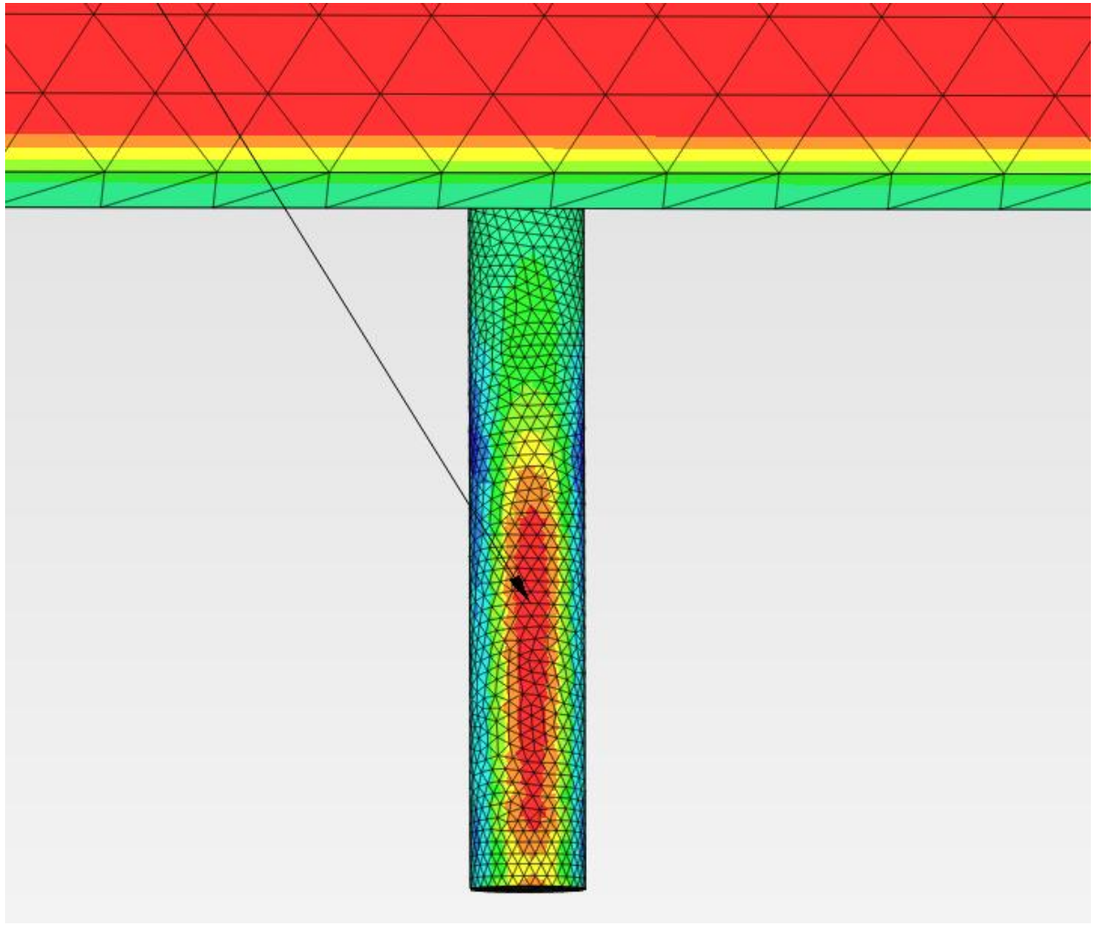
Importing Pressure Results from OpenFOAM to PrePoMax

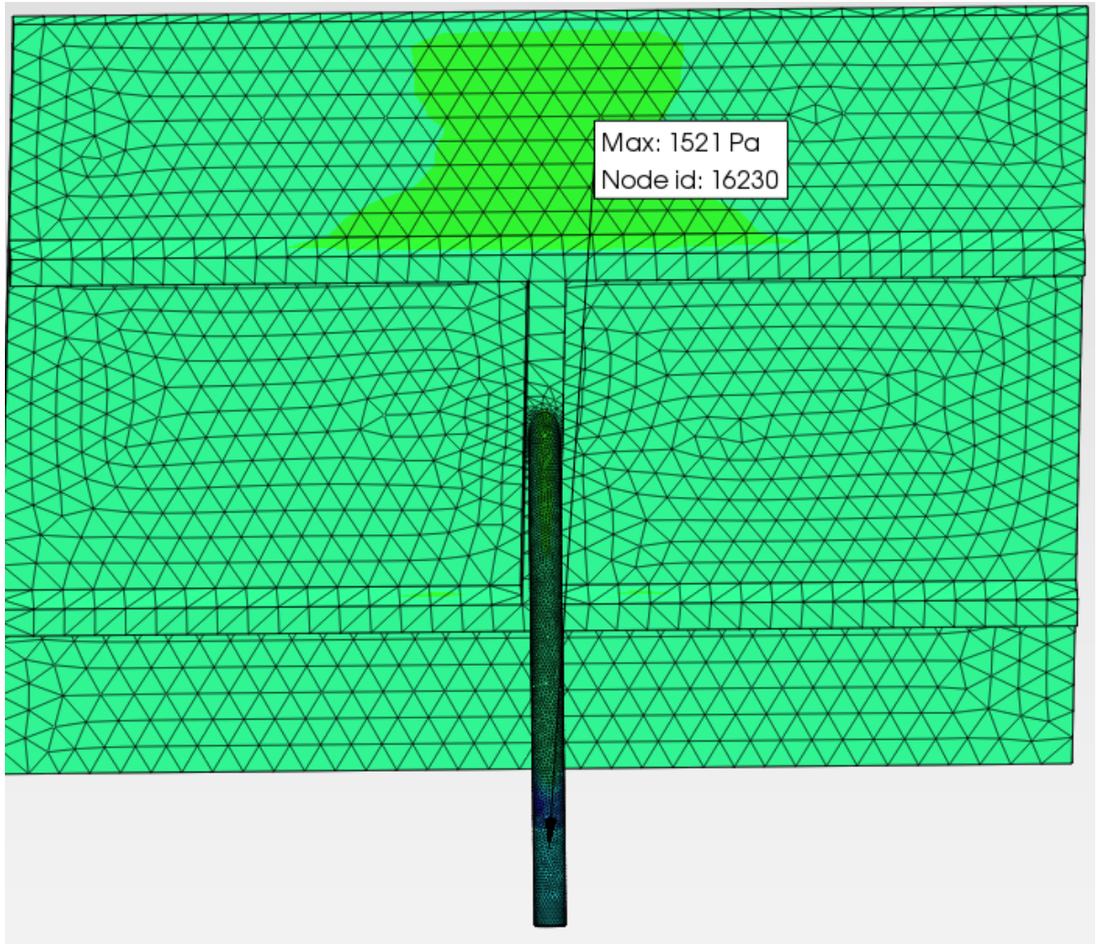




Structural FEM Simulation

Importing Pressure Results from OpenFOAM to PrePoMax

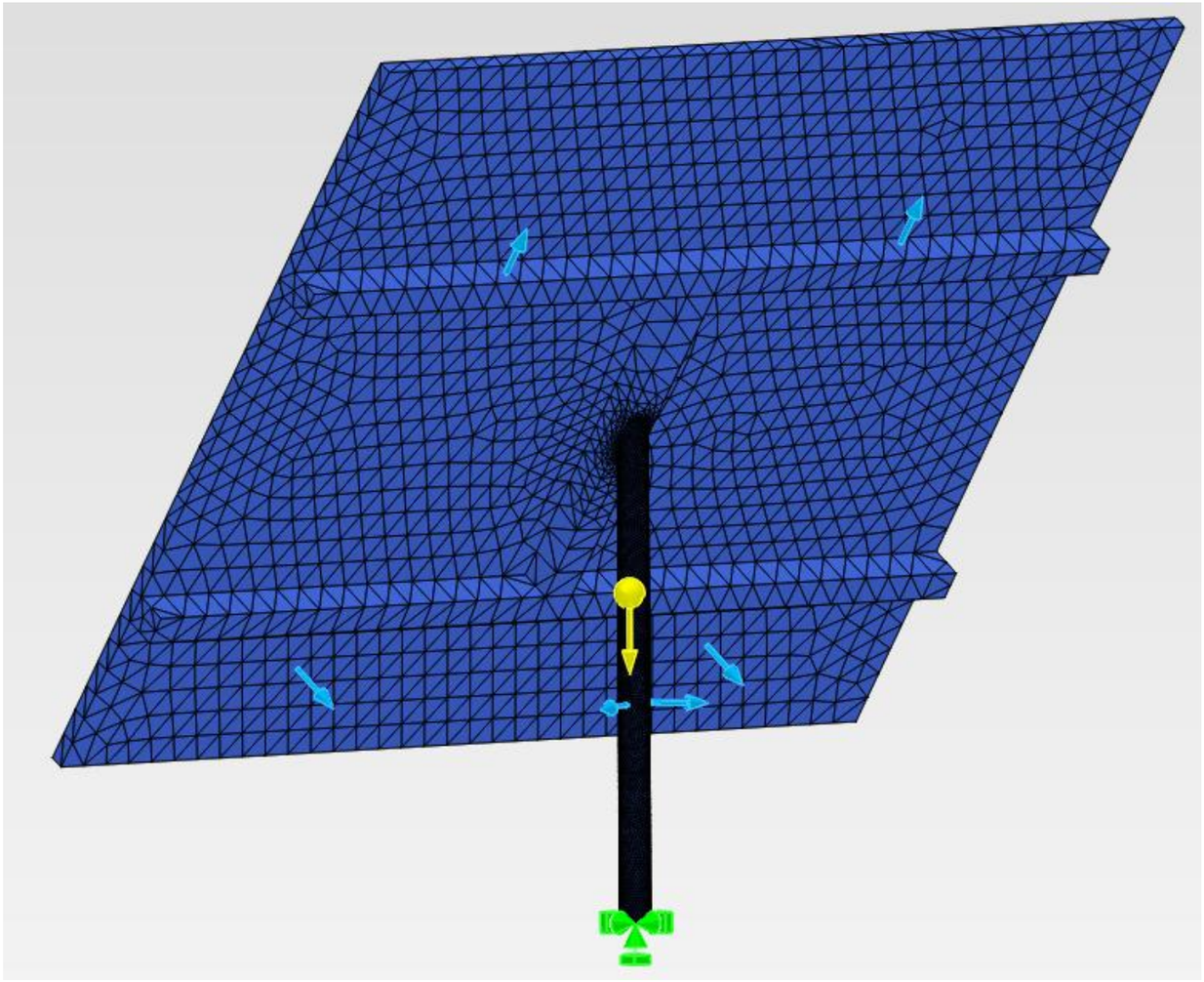




Structural FEM Simulation

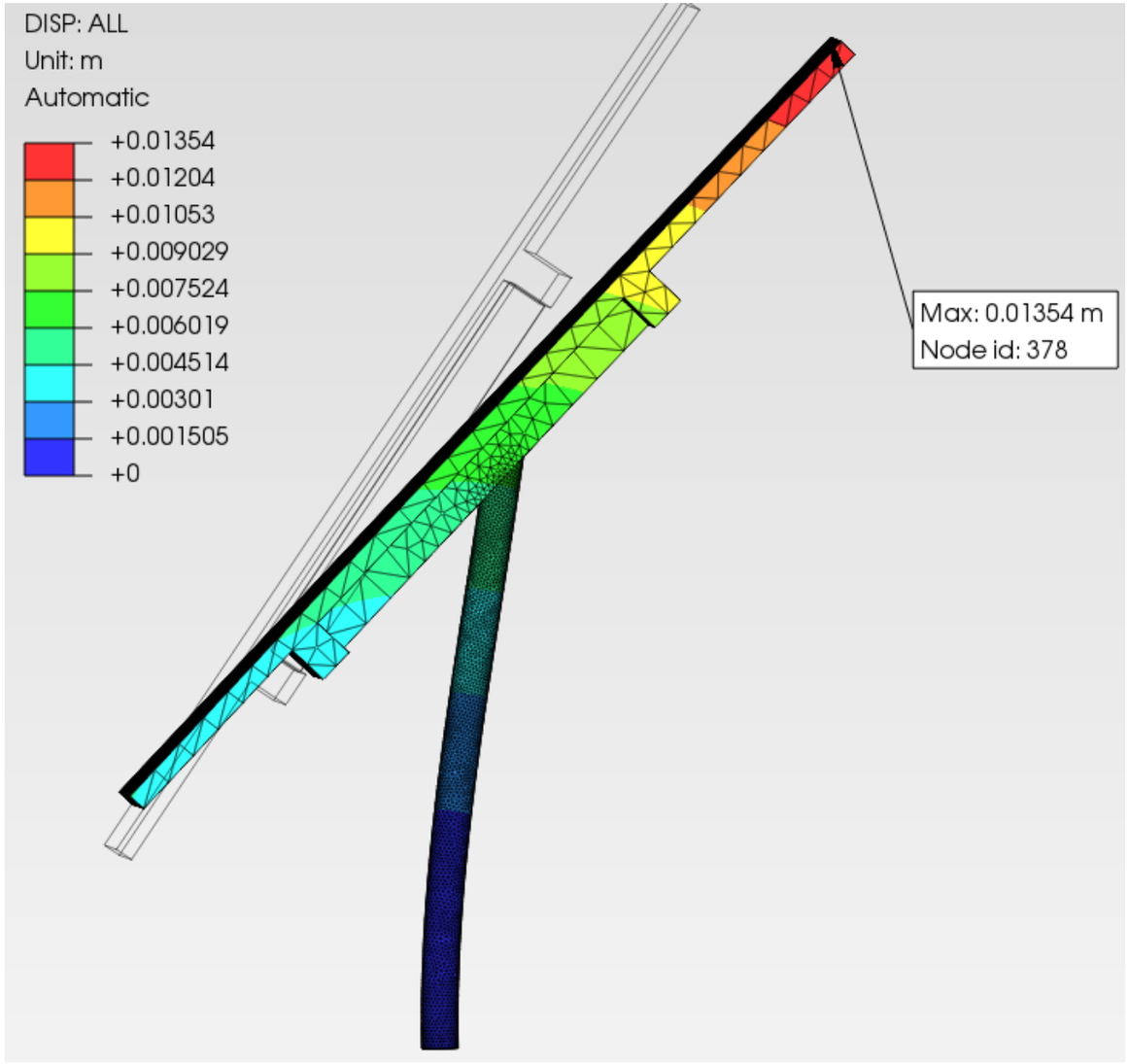
Remaining Boundary Conditions:

- Gravity – yellow
- Fixed support – green
- Imported pressure – blue



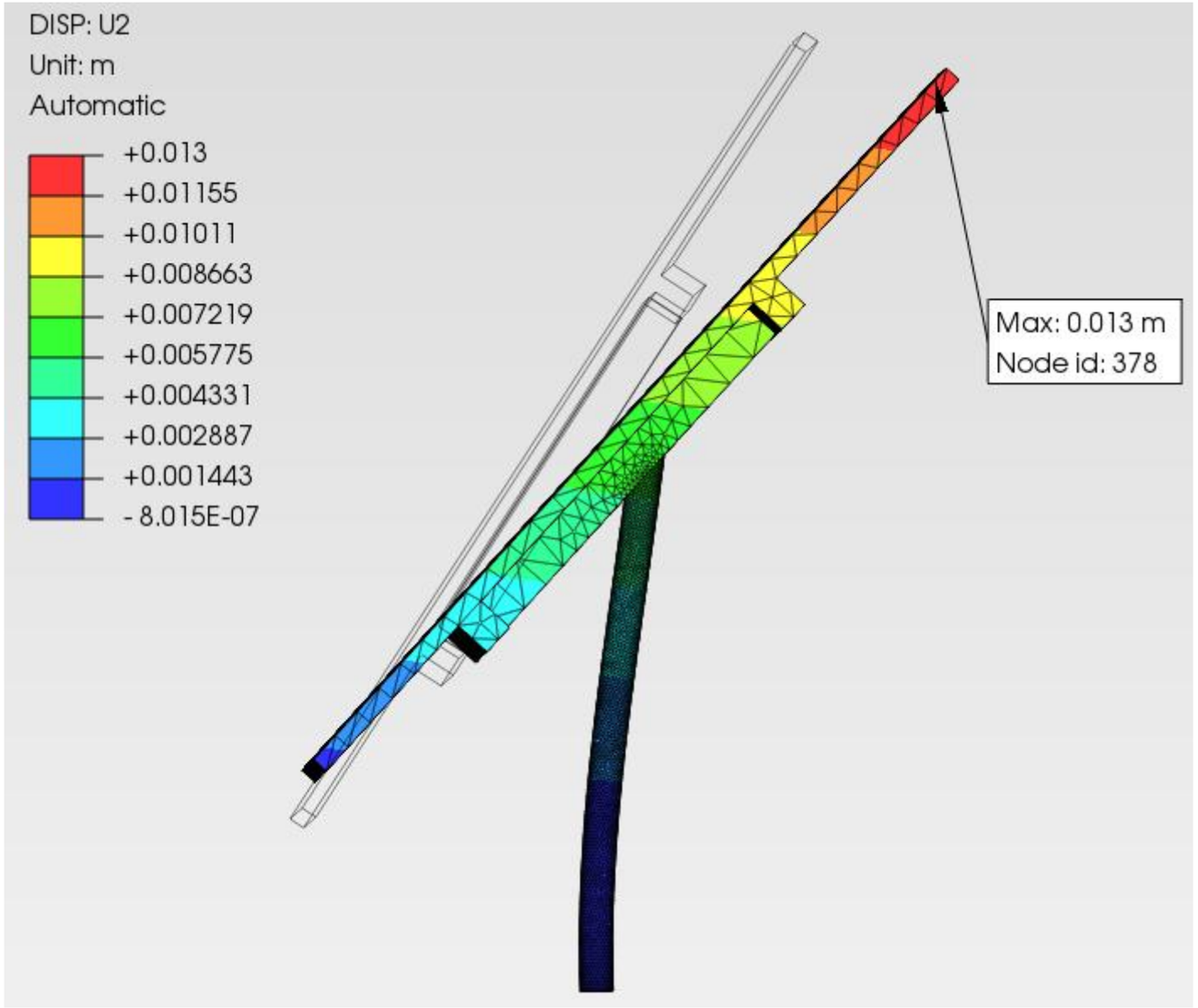
Results

Total Displacements



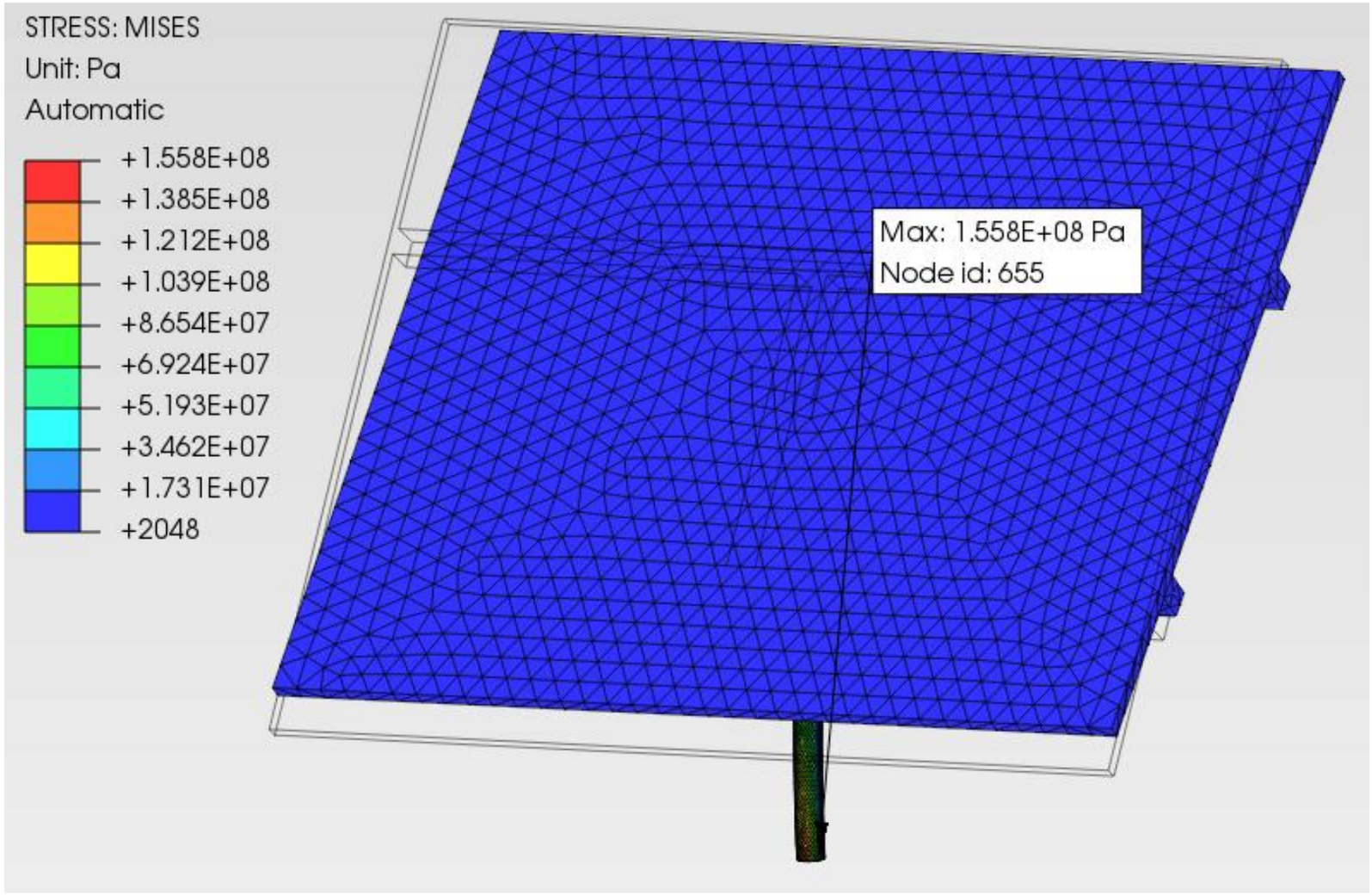
Results

Displacements in Y-Axis



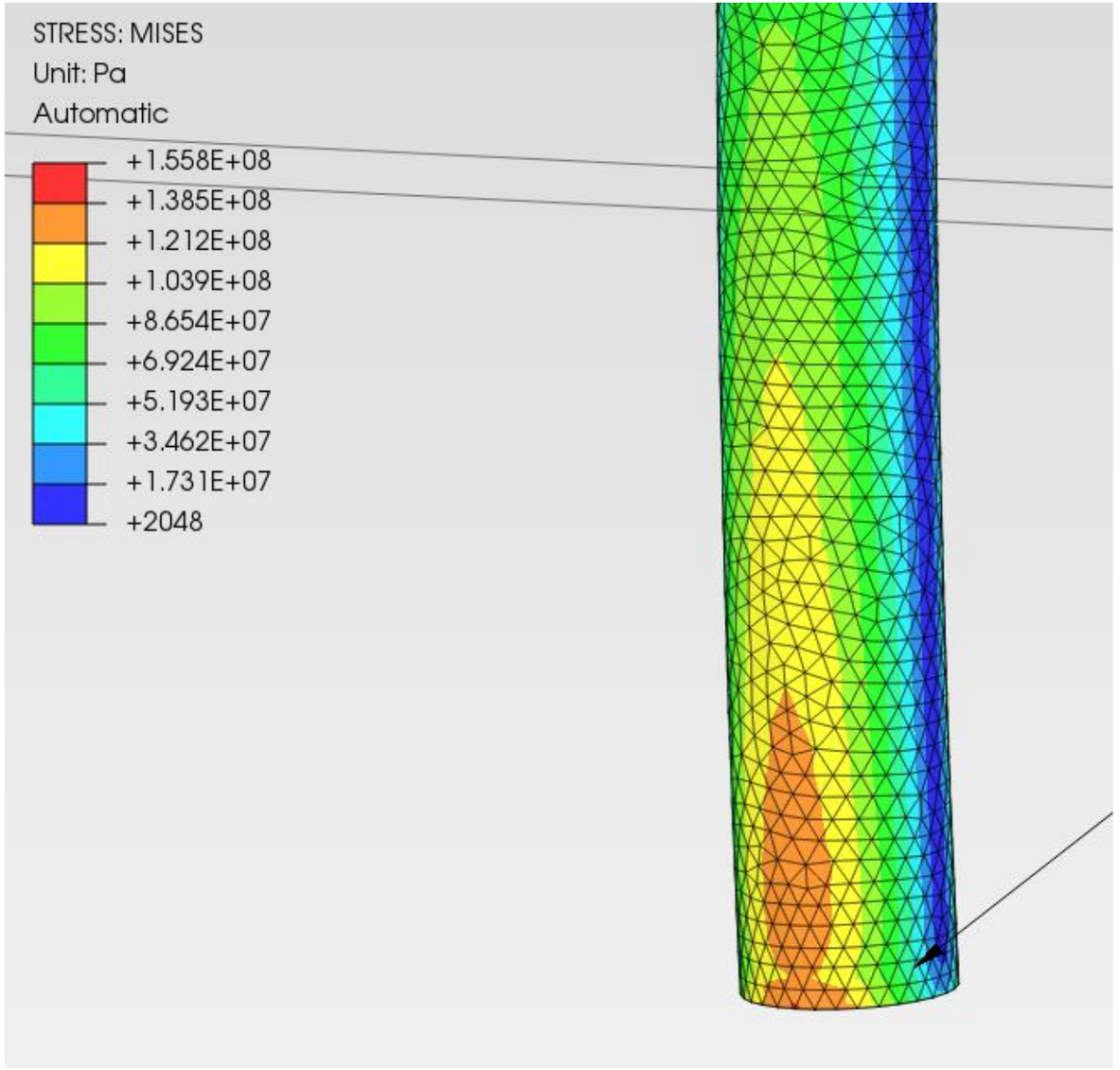
Results

Equivalent (Von Mises) Stresses



Results

Equivalent (Von Mises) Stresses



Conclusions

1. Using OpenFOAM, the wind-induced load on the structure was successfully simulated.
2. With the CalculiX solver in the PrePoMax environment, results were obtained to verify the structural strength criteria:
    2.1 Displacements of approx. 13 mm were considered small and thus do not negatively affect the structure’s functionality
    2.2 Maximum stresses reached approx. 150 MPa, which is about 50% of the yield strength limit

Summary

- Open-source simulations are an interesting alternative to commercial solutions
- They have a significantly higher ,,entry threshold”
- They offer many possibilities due to direct access and control over input files